

Muhammad Sayyedain

Junior AI / GenAI Engineer • Python Backend • RAG Applications

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SUMMARY

BSCS graduate focused on **AI/GenAI backend engineering** with hands-on experience building **RAG pipelines**, **LangGraph agent workflows**, **FastAPI services**, and **LLM evaluation** systems. Coded applied GenAI labs across hybrid retrieval, RAGAS, LangSmith tracing, PEFT/LoRA fine-tuning, FLAN-T5 fine-tuning, quantization, and deployment. Currently building **CACE**, a full-stack AI customs document review system using Python, FastAPI, LangGraph, RAG, SQL/Supabase, Docker, and Flutter.

TECHNICAL SKILLS

GenAI / LLM Engineering: RAG, LangChain, LangGraph, ReAct agents, tool calling, prompt engineering, structured outputs, embeddings

Retrieval & Evaluation: FAISS, ChromaDB, BM25, hybrid search, RRF, cross-encoder reranking, RAGAS, LangSmith, recall@k

Fine-tuning & Optimization: Hugging Face Transformers, HF Trainer, FLAN-T5 fine-tuning, PEFT/LoRA, quantization, BitsAndBytes, ROUGE/BLEU

Backend: Python, FastAPI, REST APIs, Pydantic, async/await, SQL, PostgreSQL, MySQL, SQLite

Frontend / Mobile: Flutter, Dart, Firebase Auth, Firestore, Supabase

Tools: Git, Docker, Linux, Postman, Hugging Face, Google Colab

EXPERIENCE

OptimusFox — Lahore, Pakistan

Jun 2025 – Jul 2025

Mobile App Development Intern

- Developed a **Flutter/Firebase** property booking application with listings, bookings, favorites, admin management, and role-based user flows.
- Designed **Firestore** collections and query patterns for properties, bookings, users, favorites, and role-specific views.
- Implemented separate **Admin/User** flows with authentication state, permissions, and reusable UI/provider logic.

PROJECTS

CACE — Customs Automation & Compliance Engine (GenAI Capstone, in progress)

2026

Python, FastAPI, LangGraph, LangChain, RAG, PostgreSQL/Supabase, Flutter, Docker

- Building a full-stack AI document review system for export invoices and packing lists using a **Flutter** frontend, **FastAPI** backend, and **Supabase/PostgreSQL** storage.
- Designed a **LangGraph workflow** with extraction, validation, HS-code suggestion, compliance retrieval, Q&A, and **human-in-the-loop** review nodes.
- Implemented RAG over customs/reference documents to ground compliance answers and flag mismatches in exporter details, declared values, weights, and HS-code information.
- Added API validation with **Pydantic**, persistent case history, reviewed-status tracking, and Dockerized backend setup for reproducible demos/deployment.

Production RAG Chatbot & Evaluation Pipeline (Crewlogix GenAI Bootcamp)

2026

Python, LangChain, ChromaDB, FAISS, BM25, RAGAS, LangSmith, FastAPI

- Built an end-to-end **RAG pipeline**: document loading, chunking, embeddings, persistent vector storage, retrieval, and grounded generation.
- Implemented **hybrid BM25 + dense retrieval** with RRF and cross-encoder reranking, plus multi-turn memory and Pakistan-specific **PII redaction** for CNIC and phone numbers.
- Evaluated retrieval and answer quality using **RAGAS** metrics including faithfulness, answer relevancy, context precision, and context recall; traced failures with **LangSmith**.

FLAN-T5 Fine-tuning & Quantization Lab (Applied GenAI Lab)

2026

Python, Hugging Face Transformers, HF Trainer, PEFT/LoRA, BitsAndBytes, ROUGE

- Fine-tuned **FLAN-T5** on an instruction-style/customer-service dataset using Hugging Face Trainer and compared pre/post outputs with **ROUGE**.
- Applied **PEFT/LoRA** concepts to reduce trainable parameters and understand fine-tuning tradeoffs under limited Colab T4 GPU resources.
- Experimented with **INT8/INT4 quantization** and BitsAndBytes to understand memory, speed, and quality tradeoffs for deployment.

PalPath — AI-Assisted Dental X-ray Analysis & Clinic Workflow Platform (Final Year Project)

2025 – 2026

Flutter, Python, FastAPI, YOLOv8, Firebase, Supabase

- Built a Flutter-based clinic workflow app with dentist/student roles, patient records, case management, appointments, and quiz module using Firebase and Supabase.
- Served a **YOLOv8** dental image detection model through a Python REST API and integrated predictions into the Flutter case-review flow.

EDUCATION

Bahria University — Lahore, Pakistan

Sep 2022 – Jun 2026

Bachelor of Science in Computer Science (BSCS)

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Databases, Operating Systems

CERTIFICATIONS & TRAINING

Crewlogix PSDF GenAI Bootcamp — The Institute of Technology

May – Jul 2026

13-week applied GenAI programme covering RAG, LangChain, LangGraph, agents, RAGAS, LangSmith, guardrails, FLAN-T5 fine-tuning, PEFT/LoRA, quantization, FastAPI, and deployment.

Supervised Machine Learning: Regression and Classification — Stanford Online (Coursera) | [Certificate](#)

IBM Prompt Engineering Basics — IBM (Coursera, 2026)